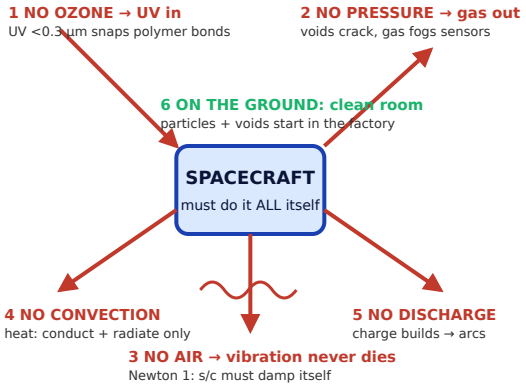
SEC 2 · LEC 4 · SPACE VACUUM CONDITIONS

Vacuum = six things Earth does for free

Six slides, one idea: on Earth the air quietly does jobs for you. In vacuum the spacecraft has to do every one of them itself. Drawn at your small-page budget.

S2 / L4 — Space vacuum **11/7 · p__**



⭐ 3 TAKEAWAYS

1. Vacuum removes 5 free services: ozone, pressure, air damping, convection, discharge
2. UV photon energy > bond energy ∴ polymers break
3. Fix most of it before launch: materials, bake-out, clean rooms

🧠 SUMMARY · video closed · 3 lines

1. _____
2. _____
3. _____

CUES **NOTES**

What does ozone block, and in what unit?

Why does UV eat polymers but not metal?

Outgassing: cause → 2 harms → 2 fixes?

Why does a vibration die on Earth but not in space?

Only 2 ways heat moves in vacuum?

Where does static go in space?

? what IS "spacecraft charging"

no ozone → UV < 0.3 µm (= 300 nm) reaches s/c
UV photon energy > C–C bond ∴ polymers go brittle/yellow (not metals)
outgassing: trapped gas in voids, 1 atm in / 0 out → cracks
escaped gas lands on cold lenses/sensors → keep outgassers away from sensors
fix: no voids in mfg · bake-out · low-outgas materials
clean room + constant testing (JUNO arrays) — 1 particle can kill a sensor
Newton 1: on Earth air absorbs vibration; in space nothing does → s/c must damp itself
no convection → heat only by conduction + radiation → hot spots stay hot
no discharge path → charge builds up → arcs; need grounding/coatings/active bleed
∴ Earth's air = free ozone, damper, fan, ground wire. Space: build all 4.

How to use this

- **Draw the center box and 5 red arrows yourself first.** Label each arrow with what's missing. Mine is for checking, not copying.
- Notes column is at its 10-line limit. Yours can be shorter.
- The ∴ line at the bottom is the one to green-highlight on your +1 day review if you can say it from memory.
- Page 2 → fact checks: two things the lecture got loose, one it got wrong.

⚡ Page 3 is the vocabulary. Read it BEFORE you copy anything: a word you can't explain is a word you can't remember.

SEC 2 · LEC 4 · TEACH + FACT CHECK

The air is doing four jobs for you right now

Plain-English version of the lecture, then where the instructor was loose or wrong. Read this on screen; it does not go in the notebook.

🌬 What Earth's air does for free

- **Sunscreen (ozone).** The ozone layer swallows the nastiest ultraviolet, everything shorter than about 0.3 µm (300 nanometers). In space that light hits the spacecraft raw. A UV photon at 300 nm carries about 4 eV, and the carbon-carbon bonds that hold plastics together are about 3.6 eV. The photon wins. Chains snap, the plastic goes brittle and yellow. Metals don't care; their bonds aren't the same kind.
- **Pressure (outgassing).** Every material made in air has tiny pockets of air inside it at one atmosphere. In vacuum the outside is zero, so that trapped gas pushes out. Two harms: the material can crack, and the escaping gas drifts off and condenses on the coldest nearby surface, which is usually a lens or sensor. That's the real reason the slide says "keep outgassing materials away from sensors." Fixes: manufacture without voids, pick low-outgassing materials, and bake the parts in a vacuum oven before launch so they outgas on the ground instead.
- **A damper (Newton's 1st law).** Flick a ruler on your desk and it stops buzzing because the air steals its energy. In space nothing steals it. A vibration, a wobble, a spin: it keeps going until the spacecraft itself kills it with dampers or reaction wheels. That's why the ADCS section exists.
- **A fan (no convection).** On Earth hot air rises and carries heat away. In vacuum there is no air to rise. Heat can only move by conduction through solid parts or by radiating into space. A hot chip stays hot unless you give it a metal path to a radiator.
- **A ground wire (no discharge).** Static on you bleeds off into the air and the floor. In space it has nowhere to go, so charge builds up on the spacecraft until it jumps as a spark between two surfaces. Fixes: conductive coatings, grounding straps between every part, sometimes an active device that spits charge away.

🔎 Fact checks

- **Wrong (in the captions):** "wavelengths smaller than 0.3 *millimeters*." The slide is right: 0.3 **micrometers** (µm), a thousand times smaller. 0.3 mm would be infrared and harmless. Write µm.
- **Loose:** "corrosion" of polymers. Corrosion is a chemical attack, usually with oxygen. UV damage is photons breaking bonds. Different mechanism, same result (brittle, cracked). In low Earth orbit the thing that really chews polymers is atomic oxygen, which *is* corrosion. Keep the two separate in your head.
- **Loose:** "no dissipation in space." Materials still have a little internal damping, so a vibration does fade eventually. It just fades far too slowly to rely on. The point stands; the word "no" is an exaggeration.
- **Simplified:** outgassing presented mainly as cracking. In practice the contamination of optics is the bigger headache; the slide's own last bullet hints at it.

📒 Question bank cards

- Ozone absorbs UV below what wavelength? (unit!)
- Why does UV damage plastics but not aluminum?
- Name two harms of outgassing and two fixes.
- A reaction wheel starts vibrating. Why doesn't it settle on its own in orbit?
- List the two heat paths that still work in vacuum.
- What's the Earth-side equivalent of a grounding strap on a spacecraft?
- Five "free services" the air provides on Earth that vacuum removes?

⚡ Next: write the 3-line summary with the video closed, then send me a photo of the spread. Then play the "atmospheric conditions" lecture.

SEC 2 · LEC 4 · VOCABULARY

Every word in this lecture, in plain English

The instructor uses these as if you already know them. You don't need to copy this page. Read it once, then on the +1 day review try to explain three of them out loud without looking.

Vacuum — space with (almost) no air in it. Not "nothing"; just no gas pushing on things. *Like the inside of a sealed jar after you pump all the air out.*

Ultraviolet (UV) — light that's a bit "bluer than blue," too high-energy for your eyes to see. It's what sunburns you. *The invisible part of sunlight that does damage.*

µm (micrometer / micron) — one millionth of a meter. 0.3 µm = 300 nm. A human hair is ~70 µm. *The caption said "millimeters": wrong by 1,000×.*

Organic chemical bond — the "glue" between carbon atoms that holds those chains together. Takes a certain amount of energy to break. *UV has more than that amount, so it snaps the glue.*

Corrosion — chemical attack on a material, like rust on iron. The instructor used it loosely for UV damage; strictly, UV damage is bond-breaking, not corrosion. *Rust = corrosion. Sun-rotted plastic = degradation.*

Void — a tiny bubble or pocket inside a material left over from manufacturing, usually full of air. *A bubble in a cake.*

Clean room — a sealed workspace with filtered air, bunny suits, and strict rules, so dust and skin flakes don't land on the spacecraft. *An operating theater for hardware.*

Dissipation — energy leaking away into the surroundings until the motion stops. *The reason a plucked guitar string goes quiet.*

Vibration / damping — a shaking back and forth / anything that makes it fade out. *Shock absorbers on a car are dampers.*

Conduction — heat moving through a solid by touch, atom to atom. *A metal spoon getting hot in soup.*

Equilibrium / "heat inequilibrium" — equilibrium = everything settled at the same temperature. Inequilibrium = a hot spot that hasn't spread out yet. *In vacuum, hot spots stay hot spots.*

Spacecraft charging — the spacecraft slowly collecting electric charge from space (sunlight knocks electrons off, charged particles stick). With no air to bleed it away, it builds until it sparks. *You on a carpet, forever, with no doorknob.*

Ozone layer — a thin band of a special oxygen (O_3) high in Earth's air that soaks up harsh sunlight before it reaches you. *Earth's sunscreen.*

Wavelength — the size of one ripple of a light wave. Shorter wavelength = more energy = more damage. UV is shorter than visible light. *Small ripples hit harder.*

Polymer — any plastic, rubber, or similar material made of long repeating chains of molecules. Kapton, Teflon, epoxies, cable insulation. *Long molecular spaghetti.*

Degradation / decomposition — the material slowly falling apart: going brittle, yellowing, cracking, flaking. *Like a garden hose left in the sun for 5 years.*

Outgassing — gas slowly leaking out of a solid material once it's in vacuum. *Like a new car's "new car smell": plastics releasing trapped stuff, but in space it lands on your camera lens.*

Pressure difference — one side pushing harder than the other. Air trapped in a void pushes out at 1 atmosphere; vacuum pushes back with nothing. *A balloon taken up a mountain swells for the same reason.*

Perturbation — any small unwanted push, nudge, or wobble. A thruster puff, a solar panel unfolding, a wheel spinning up. *Someone bumping your desk.*

Newton's 1st law (inertia) — a thing keeps doing what it's doing (moving or sitting still) unless something pushes on it. *On Earth, air and friction are the "something." In space, there isn't one.*

Convection (natural) — heat carried away by moving air or liquid: hot air rises, cool air replaces it. *Why a radiator heats a whole room.*

Radiation (heat) — heat glowing away as invisible light, needing no air at all. *Feeling a campfire on your face from a distance.* In vacuum, conduction + radiation are the only two left.

Electrical discharge — built-up charge suddenly jumping somewhere: a spark. *The zap from a doorknob after walking on carpet.*

Passive vs active system — passive = works with no power or decisions (a coating, a strap, a radiator). Active = does something on command using power (a heater, a reaction wheel, a charge-bleeding device). *A window shade vs. air conditioning.*